

AZMALINE Syrup



Composition:

Each 5 ml AZMALINE Syrup contains:

Bromhexine Hydrochloride	4 mg
Guaifenesin	50 mg
Terbutaline Sulfate	1.25 mg
Menthol	2.5 mg

Pharmacological Properties & Mechanism of Action:

Bromhexine hydrochloride: An expectorant/mucolytic, it increases the volume of sputum and reduces the viscosity of bronchial secretions, making breathing easier.

Guaifenesin: An expectorant, it works by increasing respiratory tract fluids and decreasing the viscosity of bronchial secretions and aiding in their elimination.

Terbutaline sulfate: is a selective β_2 -adrenergic agonist that dilates bronchi by relaxing bronchial smooth muscle and increases mucociliary clearance.

Menthol: gives a soothing effect to the irritated mucous membrane of the pharynx.

Pharmacokinetics:

Bromhexine hydrochloride: It is rapidly absorbed from the gastrointestinal tract. It is highly bound to plasma proteins. About (85-90%) of the dose is excreted in the urine mainly in the form of metabolites.

Guaifenesin: It is well absorbed from the gastrointestinal tract, undergoes metabolism and is excreted in the urine.

Terbutaline Sulfate: Terbutaline sulfate is absorbed variably from the gastrointestinal tract. It undergoes extensive metabolism during the first hepatic pass. It has no known active metabolites. The half-life of terbutaline sulfate is approximately 3-4 hours. Part is excreted in the urine as inactive metabolites and part as unchanged terbutaline.

Indications:

AZMALINE has an expectorant, bronchodilator and mucolytic effect. It is used in the treatment of productive cough when associated with bronchospasm in conditions such as bronchitis, bronchial asthma, chronic obstructive pulmonary disease (COPD) and abnormal bronchiectasis (bronchial dilation) and emphysema.

Contraindications:

- Hypersensitivity to the active substances, to other sympathomimetic amines or to any of the excipients.
- AZMALINE should not be used in patients with pre-existing ischemic heart disease, or in patients with significant risk factors for ischemic heart disease.

Warnings & Precautions:

- Caution should be observed while prescribing AZMALINE to patients with hypertension, cardiovascular disease (including arrhythmias, coronary insufficiency), uncontrolled diabetes mellitus, and patients with hyperthyroidism, history of seizures, or in patients who are unusually responsive to sympathomimetic amines.

- Patients at risk of hypokalemia should be monitored because transient decreases in serum potassium levels have been reported with concomitant use of beta-agonists.

- Since mucolytics, such as Bromhexine, may affect the gastric mucosa Bromhexine should be used with care in patients with a history of peptic ulceration.

- Hypersensitivity reactions, including severe skin reactions, have been reported with bromhexine. If they occur, treatment should be stopped immediately.

Pregnancy & Lactation:

Safety of AZMALINE has not been studied in pregnancy and lactation. Therefore, probable benefits should be weighed against possible risks, before prescribing AZMALINE during pregnancy and lactation.

Drug Interactions

- Hypokalemia resulting from high doses of β_2 agonists may result in increased susceptibility to digitalis induced cardiac arrhythmias.

- Hypokalemia may be enhanced by concomitant administration of aminophylline or other xanthines, corticosteroids or by giving diuretics.

- Terbutaline reduces serum theophylline concentrations by increasing its systemic clearance.

- Other sympathomimetic bronchodilators and epinephrine should not be used concomitantly with terbutaline since their combined effect on the cardiovascular system may be deleterious to the patient.

- Terbutaline should be administered with caution in patients being treated with monoamine oxidase (MAOIs) inhibitors or tricyclic antidepressants.

Side Effects:

The most frequently observed side effects are tremor and nervousness. Other commonly reported effects include increased heart rate, palpitations and dizziness. Other side effects include headache, drowsiness, vomiting, nausea, sweating, muscle cramps, hypersensitivity, rash, shortness of breath, anxiety and dry mouth. These effects are usually transient and do not require treatment.

Driving & using machines:

The most common side effects of terbutaline are tremor, headache, and tachycardia. This may affect your ability to drive and use machines.

Dosage & Administration:

AZMALINE is usually given three times daily in the following doses:

Adults and children over 12 years: 10 ml (2 small spoon).

Children 6 to 12 years: (5 - 7.5) ml (1 - 1.5 small spoon).

Children 2 to 6 years: (2.5 - 5) ml (0.5 - 1 small spoon).

Overdose:

Symptoms: Taking large doses of guaifenesin may cause nausea and vomiting. Taking large doses of terbutaline causes irregular heartbeat, palpitations, insomnia, anxiety, muscle spasms, and obvious tremor. Gastrointestinal complications, including nausea and vomiting, may occur. Sometimes hypokalemia, hyperglycemia and lactic acidosis, hypotension occur.

Treatment: In severe cases: gastric lavage and administration of activated charcoal. Monitor acid-base balance, blood sugar, and electrolytes, especially serum potassium levels. Monitor heart rate, pulse and blood pressure. Metabolic changes must be corrected. Cardiac selective beta-blockers (such as metoprolol) may be used with caution to treat arrhythmias that cause haemodynamic deterioration.

Storage:

Keep out of reach of children. Store below 30° C.

Packaging:

Each AZMALINE carton box contains amber glass bottle of 100 ml.

-A medication is a product which affects your health, and its consumption contrary to instructions is dangerous for you.
- Follow strictly the doctor's prescription, the method of use and the instructions of the pharmacist who sold the medication.
- The doctor and the pharmacist experts in medicine, its benefits and risks.
- Do not repeat the same prescription without consulting your doctor.
- KEEP THE MEDICATIONS OUT OF REACH OF CHILDREN.
- Council of Arab Health Ministers' Union of Arab Pharmacists

Revva Pharmaceutical Industry - Syria

